

Prioritizing projects in rural electrification planning: Integrating MAUT with OnSSET

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ABSTRACT

This study employs the Open-Source Spatial Electrification Tool (OnSSET) to identify the least-cost technology options, investment requirements, and capacity needed to electrify all unelectrified communities in Sokoto State, Nigeria. Based on OnSSET modelling, the total number of unelectrified settlements in the study area is 1,453, with a corresponding population of 1.28 million. Our results show that mini-grid PV is the least-cost electrification option for electrifying about 90% of the unelectrified settlements. We estimate the cost of providing basic electricity access to unelectrified communities at US\$231.4 million, and the additional capacity required is 125 MW. Given the large number of unelectrified settlements, and the financial requirements for achieving universal electricity access, we proceed to prioritize the settlements using the Multi-Attribute Utility Theory (MAUT). This approach identifies settlements that have the highest socio-economic benefits if electrified, based on five criteria: financial, economic, social, demographic and political. We conclude by noting that the technical, institutional, and governance constraints to achieving universal electrification in Nigeria are being addressed; however, the financial constraints persist. We recommend that financial support by multilateral development organizations should also include support for setting up assembling and/or manufacturing facilities for renewable energy equipment in Nigeria as this will substantially reduce the cost of mini-grid projects as well as the associated risks of such projects in terms of cost recovery.

List of Abbreviations

Abbreviation	Meaning
AHP	Analytical Hierarchy Process
ANP	Analytical Network Process
GIS	Geographic Information Systems
GRID3	Geo-Referenced Infrastructure and Demographic Data for Development
IEA	International Energy Agency
LCOE	Lowest Levelized Cost of Electricity
LGA	Local Government Areas
MAUT	Multi-Attribute Utility Theory
MCDA	Multi-Criteria Decision Analysis
MCDM	Multiple Criteria Decisions Making
MUF	Marginal Utility Functions.
NEP	Nigeria Electrification Project.
NREEEP	Nigeria National Renewable Energy and Energy Efficiency Policy
OnSSET	Open-Source Spatial Electrification Tool

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Abbreviation	Meaning
PROMETHEE	Preference Ranking Organization Method for Enriched Evaluation
REM	Reference Electrification Model
RESIP	Rural Electrification Strategy and Implementation Plan
SDG7	Sustainable Development Goal 7.
TOPSIS	Technique of Order Preference Similarity to the Ideal Solution
UNDP	United Nations Development Programme.

1. Background

One of the main energy sector challenges facing countries in sub-Saharan Africa is achieving universal electricity access in line with the targets of the Sustainable Development Goal 7 (SDG7). The International Energy Agency (IEA) estimated that about 675 million people in

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the world were without access to electricity in 2021, out of which 567 million (or 84%) were in sub-Saharan Africa [1]. Most of these people are in rural areas [2] which are often far from the existing grid and have relatively low energy demand [3]; [4]. Some of these rural areas have renewable energy resources which can be harnessed for off-grid electrification solutions. The use of renewable energy technologies is critical to achieving universal electricity access in line with SDG7 target in many developing countries.

Nigeria is the country with the highest number of persons without access to electricity. In 2021, the electricity access rate in Nigeria was estimated at 60% (i.e. 89% in urban areas and 26% in rural areas) [1]. This implies that about 86 million people in Nigeria did not have access to electricity [1]. The national electricity access rate in the country does not clearly reflect the disparity in the electricity access rates across the different states - which is below 30% in some states and above 90% in others [5]; [6]. The reasons for this disparity include the difference in population density and settlement patterns across the states, the coverage of the existing grid, the locations of energy resources used for grid-power generation, and prevailing income levels, among others [5]. The Government of Nigeria, through the Rural Electrification Strategy and Implementation Plan (RESIP), set the target of achieving 90% electricity access by 2030, of which 10% will be from renewable energy sources [7]. Furthermore, one of the objectives of Nigeria's National Renewable Energy and Energy Efficiency Policy (NREEEP) is "to guarantee adequate, reliable, affordable, equitable and sustainable supply of renewable energy at appropriate costs and in an environmentally friendly manner to the various sectors of the economy for national development" [8].

Renewable energy resources differ significantly across different parts of Nigeria. For example, the average daily solar horizontal irradiation hovers around 4.2 kWh/m²/day in the southern part of the country but exceeds 6 kWh/m²/day in some areas in the northern part. Similarly, the average wind energy potential in the southern and northern parts of the country is estimated at 3.0–3.5 m/s and 4.0–7.5 m/s respectively at 10 meters above the ground [9]. Achieving the electricity access targets will require the utilization of local renewable energy resources for off-grid technology options to provide electricity access to thousands of unelectrified communities. This will also take into consideration other underlying factors that affect electricity access, such as settlement patterns, population, local economic opportunities, etc. [5]; [10].

Several studies have examined the strategies for deploying renewable energy technologies to achieve universal electricity access in Nigeria [11]. The advancement in geospatial technologies has led to the integration of geographic information systems (GIS) into the process of determining the feasibility of deploying renewable energy technologies for electricity generation in different configurations, such as mini-grid or standalone systems. This has led to the development of several tools used for assessment and decision support. Some of these tools include Network Planner, Reference Electrification Model (REM), and the Open Source Spatial Electrification Tool (OnSSET). Studies have applied these tools in different countries in sub-Saharan Africa including Ghana [12], Malawi [13], Tanzania [14]; [15], Ethiopia [16] Nigeria [5]; [17–21]. Isihak et al. [20] reviewed the tools that are used for GIS-based rural electrification planning and noted that OnSSET is the most versatile because it builds on the methods developed by the earlier tools, and integrates several spatial and socio-economic variables that are relevant for rural electrification planning.

The results from OnSSET modelling usually provide the list of unelectrified communities alongside several other attributes, such as the least-cost option for electrification, appropriate technology to deploy, required investment cost, population that will be electrified, capacity required in kW, among others. The OnSSET result is one of the key steps in the planning process for achieving universal electricity access and can form part of a multi-year electrification strategic plan of a country or sub-national region. The decision to invest in electrifying communities goes beyond this step: it requires selecting specific

communities—among thousands—that should be prioritized. This is particularly crucial for developing countries, which often face significant budgetary, technical, and institutional constraints in delivering electricity access projects. Prioritization ensures that communities with the highest potential socio-economic benefits are electrified first, leading to a greater multiplier effect on the overall economy, and accelerating progress toward achieving SDG7 targets.

The process of prioritizing communities for electrification is complex, as it involves more than just comparing investment costs. It requires careful consideration of the unique characteristics of each community. For example, if grid-extension is being considered, the distance to the existing grid varies across unelectrified communities. Similarly, if off-grid solutions are being explored, factors such as the availability of energy resources, socioeconomic activities, and demographic characteristics differ from one community to another, making tailored solutions essential. There are also political considerations in the provision of electricity access, especially in developing countries like Nigeria where electrification of rural communities may be used for political purposes. Given the diverse and often conflicting criteria involved in the decision-making process to select communities, multi-criteria decision analysis (MCDA) has emerged as a widely used approach among researchers to support such decisions. The MCDA refers to tools and methods developed by researchers over time to support decision makers in making decisions that involve finding a compromise between competing and conflicting objectives and criteria. Kumar et al. [22] defines multiple criteria decisions making (MCDM) as a "branch of operational research dealing with finding optimal results in complex scenarios including various indicators, conflicting objectives and criteria".

There are different methods within the MCDA, and each method is appropriate to specific types of decision-making problems. The most popular of these methods are the Analytical Hierarchy Process (AHP), Analytical Network Process (ANP), Multi-Attribute Utility Theory (MAUT), Technique of Order Preference Similarity to the Ideal Solution (TOPSIS), Preference Ranking Organization METHod for Enriched Evaluation (PROMETHEE), among others. In view of the wide applicability of MCDA, researchers have built on these basic models to develop new models that are more mathematically complex. Different studies have applied these MCDA methods to answer different questions related to electricity access using renewable energy. We may categorize these studies into different strands. The first strand includes studies that have developed and/or applied customized tools [23]; [24]; the second strand includes studies focused on identifying the most suitable technology for providing electricity access to communities based on various criteria [25–28]; the third strand includes studies that focused on the best location for siting an electricity generation facility [29]; [30]; while the fourth strand includes literature review studies [31]; [22,32–34].

There is also a strand of literature that applies multi-criteria decision methods to prioritizing communities for rural electrification to achieve optimal socio-economic impacts. In their studies, Paredes-Budon & Reyes-Chamorro [35] prioritized 98 communities in the Los Rios Region in Chile using several criteria, including demographic, socioeconomic, social, productivity, and energy potential. The study by Gómez-Hernández et al. [36] prioritized 8 communities of the municipality of Cintalapa-Chiapas in Mexico and combined the analytical hierarchy process with discrete compromise programming to derive weights for different criteria used in the prioritization process. To the best of our knowledge, there is none of such studies on Nigeria despite several previous GIS-based rural electrification planning studies in the country which identify the least-cost options for electrifying communities. Hence, this study seeks to fill in this knowledge gap. It is important to note that our study is different from previous studies that have applied two-stage approach for prioritizing electrification in Nigeria [37]; [38]. These studies applied multi-criteria decision methods to select the optimum electrification design for different communities in Nigeria while our study aims to prioritize communities for

electrification after the least-cost technology option for each community has been identified. Specifically, the objectives of this study are:

- (i) to determine the least-cost technology options, investment requirement, and capacity needed for electrifying all unelectrified communities in Sokoto State, Nigeria using OnSSET; and
- (ii) to prioritize these unelectrified communities for electrification projects in order to maximize socio-economic benefits using MAUT.

The remaining part of this study is organized as follows: following this introduction, we will present the methodologies used in this study. We will start by presenting the OnSSET method, and thereafter, the method for MAUT. The third segment will be the results and discussions, while the fourth segment will be the conclusion and policy implications.

2. Methodology

2.1. Study location

This study focuses on Sokoto State, Nigeria, selected primarily due to data availability and its rich renewable energy potential, including micro-hydroelectric resources and high solar irradiation. Sokoto, located in northwestern Nigeria, is one of the country's 36 states. It shares borders with Niger Republic to the north, Kebbi State to the west and south, and Zamfara State to the east, as illustrated in Fig. 1. The state has a landmass of about 25,973 square kilometres and a population of 6.3 million in 2022. Based on the most recent state-level data, the electricity access rate of the state was estimated at 38.9% in 2013 [39].

2.2. Open Source Spatial Electrification Tool (OnSSET)

OnSSET is a GIS-based rural electrification planning tool developed by the KTH Royal Institute of Technology in Sweden. It serves as a decision-support tool for achieving SDG7. The development of OnSSET is sponsored by several international organizations, including the United Nations Development Programme (UNDP), the World Bank, and the International Energy Agency (IEA), among others.

The tool utilizes various spatial and non-spatial datasets related to rural electrification planning to identify the least-cost technology options for achieving electricity access targets at national or subnational levels. Assessments are conducted at the most granular level, such as hamlets or communities. OnSSET is written in Python programming language, and typically runs on the Anaconda environment. The summarized step-by-step process for running the OnSSET model is outlined below.

- (i) **Setting up the OnSSET runtime environment in Anaconda:** This step involves installing Anaconda and using the Anaconda command prompt to install the necessary Python packages required for OnSSET modelling. Proper configuration of the environment ensures seamless execution of the model.
- (ii) **Processing GIS files for various variables:** The OnSSET runtime environment allows users to upload multiple GIS files in different formats relevant to electrification planning. The application processes these files and extracts the necessary information into a .csv file. The list of required files, their formats, and sources is provided in Table 1.
- (iii) **Determining the least-cost technology option:** In this step, OnSSET evaluates the least-cost electrification solution from a range of options, including grid extension and off-grid solutions. For this study, the off-grid options considered were standalone PV, mini-grid PV, and mini-grid hydro, while wind and diesel options—whether standalone or hybrid—were excluded. Key inputs, such as electricity demand levels, socio-economic and

demographic data, and cost-related parameters for both grid extension and off-grid solutions, are incorporated into the model. The least-cost option for each community is determined based on the lowest levelized cost of electricity (LCOE) among the available technologies. This step generates a .csv file containing the results. The data used in this analysis are included as supplementary information.

- (iv) **Results and summaries:** The .csv file generated in the previous step provides a comprehensive list of unelectrified communities, along with relevant attributes such as geographic coordinates, population, least-cost technology option, required capacity (in kW), and investment costs (in US dollars). The data can be further analyzed using a GIS software to create maps, while additional insights can be presented as tables and charts to aid decision-making.

The detailed methodology of OnSSET is summarized in Mentis et al. [40], with the latest version available on the OnSSET website.¹ This study adopts 2020 as the base year and 2030 as the target year. While scenario analyses are typically conducted with OnSSET to assess how variations in key variables—such as technology costs and electricity demand—affect model outcomes, the primary focus of this study is to build upon the OnSSET results.

Specifically, this study aims to demonstrate how decision-makers in resource-constrained environments can prioritize communities for electrification to maximize socio-economic impact using a multi-criteria decision-making approach. Consequently, the next stage applies the Multi-Attribute Utility Theory to refine electrification priorities.

2.3. Multi-attribute utility theory

The second objective of this study is to prioritize unelectrified communities for electrification to maximize socio-economic benefits, while considering financial and institutional constraints in deploying renewable energy technologies. This is achieved using Multi-Attribute Utility Theory (MAUT), a method well-suited for prioritization when dealing with a large number of elements [41]. To do this, the process begins with defining the performance matrix.

2.3.1. Performance matrix

We define A as a set of m alternatives such that $A = \{a_1, a_2, \dots, a_m\}$. We also define C as set of n criteria to be used in assessing the different alternatives in A , such that $C = \{c_1, c_2, \dots, c_n\}$. The performance matrix, P , is the $m \times n$ matrix which presents the “raw” performance of each alternative in each criterion. For our study, the alternatives are the different unelectrified communities that have been identified using OnSSET. We use the following criteria for each community (the source of data are included in parenthesis).

- **Financial:** cost of electrification (OnSSET result);
- **Demographic:** Population (OnSSET result);
- **Social:** health - number of health centres within 2,000 meter radius (Geo-Referenced Infrastructure and Demographic Data for Development (GRID3))
- **Social:** education - number of primary schools within 2,000 meter radius (GRID3)
- **Economic:** number of markets² within 2,000 meter radius (GRID3); and

¹ <http://www.onsset.org/about.html>.

² A rural market in Nigeria is a physical marketplace where people go to buy and sell food items and general merchandize. Rural markets play a critical role in creating value, generating income for operators, and contributing to the revenue of local authorities. The number of markets in a community reflects the level of economic activities of such community.

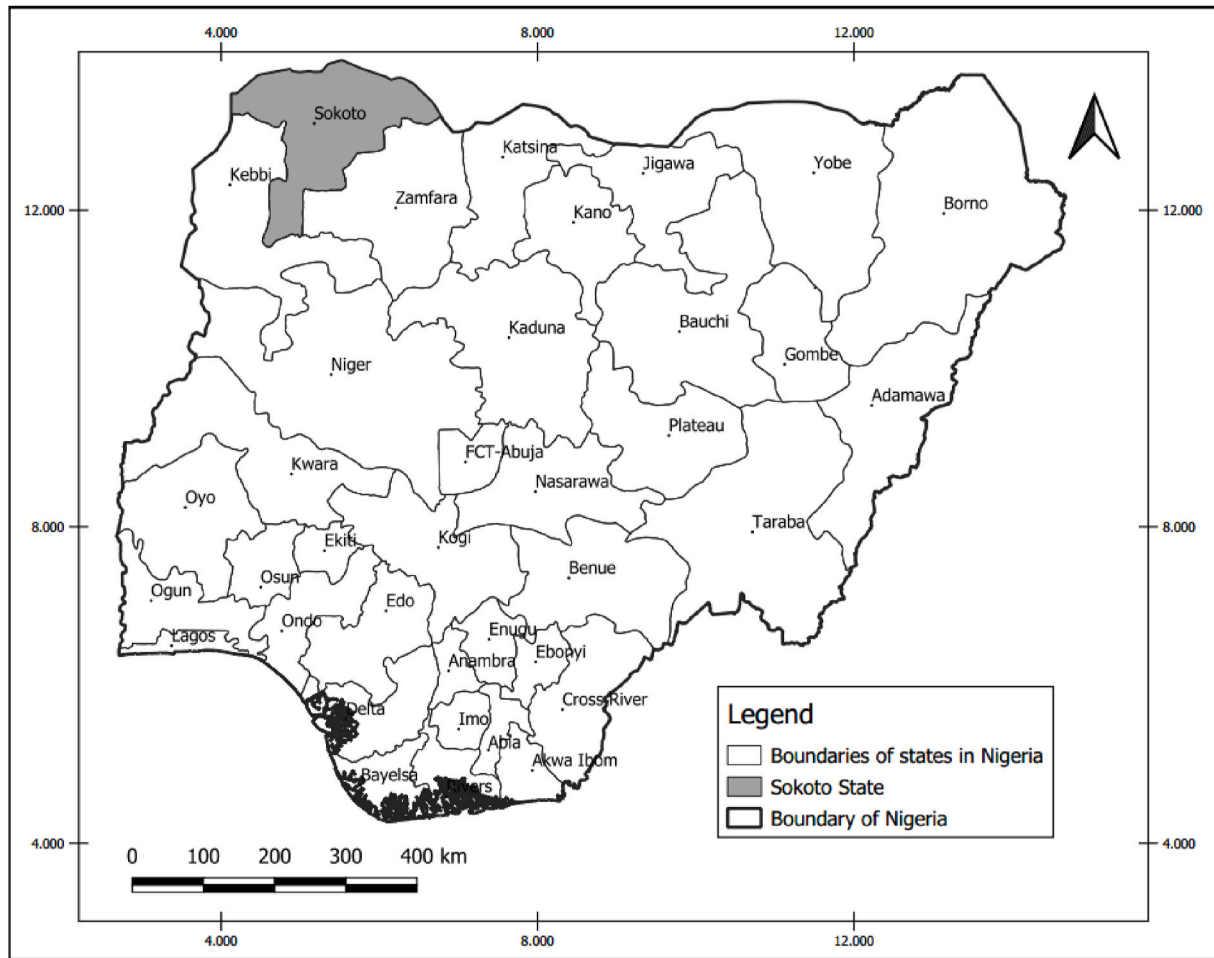


Fig. 1. Outline map of Nigeria showing Sokoto State.

- **Political:** number of election polling units (voting centres)³ within 2,000 meter radius (Independent National Electoral Commission).

The choice of the criteria is based on literature [41]. We note that some of the criteria relevant for prioritizing communities, such as energy resources availability, have already been used in OnSSET to determine the least-cost technology option as well as the estimated cost of electrification for each community. The data on social, economic, and political criteria are incorporated into the OnSSET output (.csv file) as follows: first, we upload the point vector shapefiles of the health centres, schools, markets, and polling units to QGIS as well as the .csv OnSSET output file. Second, the OnSSET output file is then converted to a shapefile, re-projected to an appropriate coordinate referencing system (i.e. EPSG:32632 - WGS 84/UTM zone 32N). Thereafter, a buffer operation using a radius of 2000 meters is carried out to convert the point shapefile to a multi-polygon shapefile. Third, the number of points (i.e. the social, economic, and political) in the polygon is obtained using the “count point in polygon” tool. Other necessary vector operations to support this process are also carried out. The summary of the data in the Performance Matrix is presented in Table 2.

³ In Nigeria, the selection of communities to electrify are usually done by political actors who have political interests are expected to include some political considerations in the decision making process. Therefore, it is important to include the political criterion. The number of polling units in an area reflects the voting power of that area and could be used to represent the political criterion.

2.3.2. Normalization of performance matrix

The units of measurements of the criteria are seldom the same, and therefore not comparable. For example, from the criteria highlighted above, we may notice that “number of persons” and “number of schools” are entirely different units. To address this, we normalize the table to remove the units and scales to make the values comparable. We do this by scaling all the values between 0 and 1, using linear scaling as follows:

For criteria where the desired outcome needs to be maximized:

$$N_{ij} = \frac{c_j(a_i) - \text{Min}_j[c_j(a_i)]}{\text{Max}_j[c_j(a_i)] - \text{Min}_j[c_j(a_i)]}$$

For criteria where the desired outcome needs to be minimized:

$$N_{ij} = \frac{\text{Max}_j[c_j(a_i)] - c_j(a_i)}{\text{Max}_j[c_j(a_i)] - \text{Min}_j[c_j(a_i)]}$$

Note that $\text{Min}_j[c_j(a_i)]$ and $\text{Max}_j[c_j(a_i)]$ represent that minimum and maximum values recorded in each column c_j respectively, and N_{ij} s are the normalized values of the performance matrix

2.3.3. Marginal utility function

The marginal utility function is an important aspect of the MAUT. It represents that amount of utility that is expected to be derived from the normalized values of the performance matrix as the values increase or decrease, and are usually scaled from 0 to 1. Some studies have specified different non-linear marginal utility functions [41]. In this study, we specify linear marginal utility functions for all criteria as this simplifies the computation. Let U_{ij} be the marginal utility that will be derived by

Table 1

GIS files and filetypes used in OnSSET analysis.

Data type	File type	Source
Administrative boundary	Vector polygon	GADM ^a
Population clusters	Geopackage/ multipolygon vector	Processed from World Pop ^b data
Population distribution layer	Raster	World Pop
Night time light	Raster	Earth Observation Group, Payne Institute for Public Policy ^c
Land cover map	Raster	US Geological Services
Elevation map	Raster	US Geological Services
Global Horizontal Irradiation (GHI)	Raster	Global solar atlas
Travel time map	Raster	European Commission Joint Research Centre
Wind velocity	Raster	Global wind atlas
Locations of substations	Point vector	Transmission Company of Nigeria
Existing high voltage lines	Line vector	Transmission Company of Nigeria
Existing medium voltage lines	Line vector	Kaduna Electricity Distribution Company
Roads	Line vector	Extracted from open street maps
Locations of distribution transformers	Point vector	Kaduna Electricity Distribution Company
Potential location of micro hydroelectric power	Geopackage/point vector	Rural Electrification Agency, Nigeria

^a <https://gadm.org/maps/NGA.html>.^b <https://www.worldpop.org/>.^c https://eogdata.mines.edu/download_dnb_composites.html.

electrifying alternative i under criteria j , where $(i = 1, \dots, m)$ and $(j = 1, \dots, n)$. The marginal utility, $U_{ij} = N_{ij}$ for all values of i and j .

2.3.4. Determination of weights

The MAUT involves assigning weights to each criterion to reflect its relative importance to the decision-maker. In this study, the Analytic Hierarchy Process (AHP) was employed to determine the criterion weights. Specifically, we use the pairwise comparison method developed by Saaty [42,43] and further elaborated by Ishizaka & Nemery [44], pp. 13–54). A total of 15 experts drawn from institutions and organizations involved in selecting communities for rural electrification projects in Nigeria participated in rating the criteria using a 9-point scale. The final weights were obtained using the arithmetic average of individual weights.

2.3.5. Determination of utility score

The utility score $U(a_i)$ for each alternative a_i is a measure of the total utility that comes from a_i after taking into consideration the marginal utility scores U_{ij} and are weights w_j of each criterion. For our study, we use the addition utility function to obtain $U(a_i)$ as follows:

$$U(a_i) = \sum_{j=1}^n U_{ij} * w_j$$

After getting the utility score for all the alternatives, we then rank them to obtain the most desirable alternatives.

Table 2

Summary of data on criteria in the Performance Matrix.

	Population (2030)	Investment Cost (2030) (US\$)	No of Polling units	No of Schools	No of Markets	No of Health facilities
Min	1	296.39	0	0	0	0
Median	333	59,733.79	1	1	0	0
Max	17359	3,237,097.21	8	5	1	3
Average	880.12	159,266.29	1.10	0.84	0.03	0.42

2.3.6. Scenario analysis

The AHP process generates weights that are applied in the MAUT analysis. However, we recognize that these weights may differ based on the preferences of decision-makers. Consequently, a scenario analysis is conducted to assess how variations in the assigned weights influence the MAUT results. Specifically, the impact of assigning higher weights to each criterion is systematically examined to understand their effects on the overall prioritization outcomes as shown in Table 3.

3. Result and discussions

3.1. Result from OnSSET analysis

The final output of OnSSET is a .csv file that contains the list of settlements, attributes of each settlement such as population, electrification status, solar energy potential, elevation, distance to the nearest road, distance to the nearest distribution line and nearest transformer, levelised cost of electrification for each technology option, least-cost technology, cost per capita, capacity required, etc. A total of 4,831 settlements are identified in the study area based on the multipolygon vector of settlement clusters as presented in Fig. 2. The coverage of the existing grid is presented in Fig. 3. OnSSET uses the coverage of existing grid and the night-time lights to predict settlements that are electrified and those that are not. Fig. 4 shows electrified and unelectrified settlements in the study area based on OnSSET's determination. We present the summary of information on population and electrification status of settlements for the base year (i.e. 2020) in Table 4.

Our analysis considered only three off-grid technologies, - i.e. standalone PV, mini-grid PV, and mini-grid hydro. Standalone diesel and mini-grid diesel are excluded because our focus is on promoting renewable energy utilization for achieving SDG7 targets in Nigeria. We present a map showing the least-cost technology options for electrifying unelectrified settlements in Fig. 5. We observe from Fig. 5 that all settlements that are compatible with mini-grid hydro are in proximity to locations with potential micro hydropower plants.

We present in Table 5 the number of settlements for which each technology option is the least-cost option. We disaggregate the number of settlements by population size to provide additional insights. From Table 5, we observe that out of the 1,453 unelectrified settlements in the study area, as identified by OnSSET, mini-grid PV is the least-cost option for 1,304 (i.e. 90%) of them in 2030. The remaining settlements have standalone PV or mini-grid hydro as the least-cost options. Among the 99 settlements compatible with standalone PV, 98 have a population of fewer than 100 people. For the 1,304 settlements compatible with mini-grid PV, 341 have populations greater than 1,000. Of the 50 settlements compatible with mini-grid hydro, 23 have populations between 100 and 500. Table 6 shows the total population of these settlements. From Table 6, we observe that the total populations for the settlements compatible with standalone PV, mini-grid PV, and mini-grid hydro are 2,379, 1.24 million, and 33,620 respectively.

Achieving universal electricity access in 2030 will require additional installed capacity. For our study, we had assumed that the initial demand level for rural households is Tier 3 demand level based on the World Bank multi-tier electricity demand levels [45]. This is because it reflects the electricity demand level that can be achieved with all the technology options used in the model in an area where there are huge constraints on financial resources to invest in systems with higher

Table 3
Weights used in scenario analysis.

Criteria	Weights in scenarios					
	Base-case	Financial	Social	Economic	Political	Demographic
Financial	0.28100	0.5	0.1	0.1	0.1	0.1
Social: Education	0.17008	0.1	0.3	0.1	0.1	0.1
Social: Health	0.29878	0.1	0.3	0.1	0.1	0.1
Demographic	0.13768	0.1	0.1	0.1	0.1	0.5
Economic: Markets	0.07178	0.1	0.1	0.5	0.1	0.1
Political	0.04068	0.1	0.1	0.1	0.5	0.1
Total	1	1	1	1	1	1

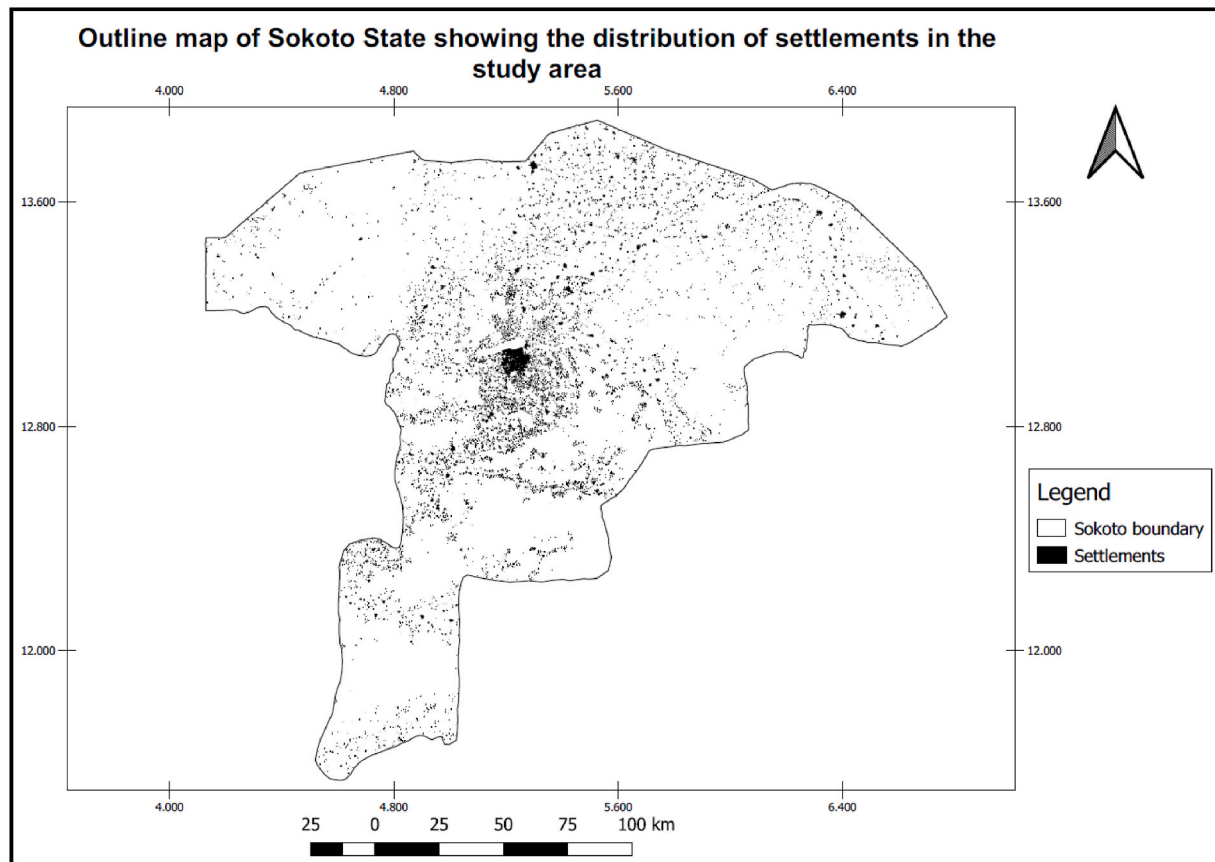


Fig. 2. Outline map of Sokoto showing the distribution of settlements (polygon vector shapefile).

capacities. We acknowledge that the use of Tier 3 electricity demand level implies usage restrictions in achieving universal electricity access.

We observe from [Table 7](#) that for settlements that are compatible with standalone PV, and whose populations are less than 100, the total additional capacity required is 198 kW. Similarly, for settlements that are compatible with mini-grid PV, - whose populations are above 1,000, the total additional capacity required to achieve universal electricity access is 92,154 kW. The investment requirement for each technology option within each population band is presented in [Table 8](#). This table shows that the total investment required to achieve universal electricity access in the study area is US\$231.41 million. The investment requirement for only the mini-grid PV is US\$227.85 million. The investments per capita for the different technology options and population bands are shown in [Table 9](#). We observe from [Table 9](#) that the per capita cost for all technologies reduces as the population increases, and that mini-grid hydro has the lowest per capita cost.

3.2. Result from multi-attribute utility theory

The first part of this study determines the least-cost technology options, investment requirements, and the generation capacity needed for electrifying all unelectrified communities in the study area using OnS-SET. While this is a necessary first step in planning for electrification investments, experience and research have shown that this is not sufficient. For example, the financial requirement for achieving this target is often beyond the reach of most countries and sub-national governments. Moreover, given the existing institutional capacity in many countries, it will be difficult to carry out electrification projects in all unelectrified communities at the same time, hence the need for prioritization. The result of the MAUT shows the utility that will be derived from electrifying each of the 4,831 communities identified in our study area. These could be arranged in a descending order for prioritization. [Table 10](#) presents the attributes of the top five unelectrified settlements prioritized for electrification projects, based on their utility scores in the base-case scenario. This scenario applies criterion weights derived using the Analytic Hierarchy Process. In this scenario, the highest utility score is

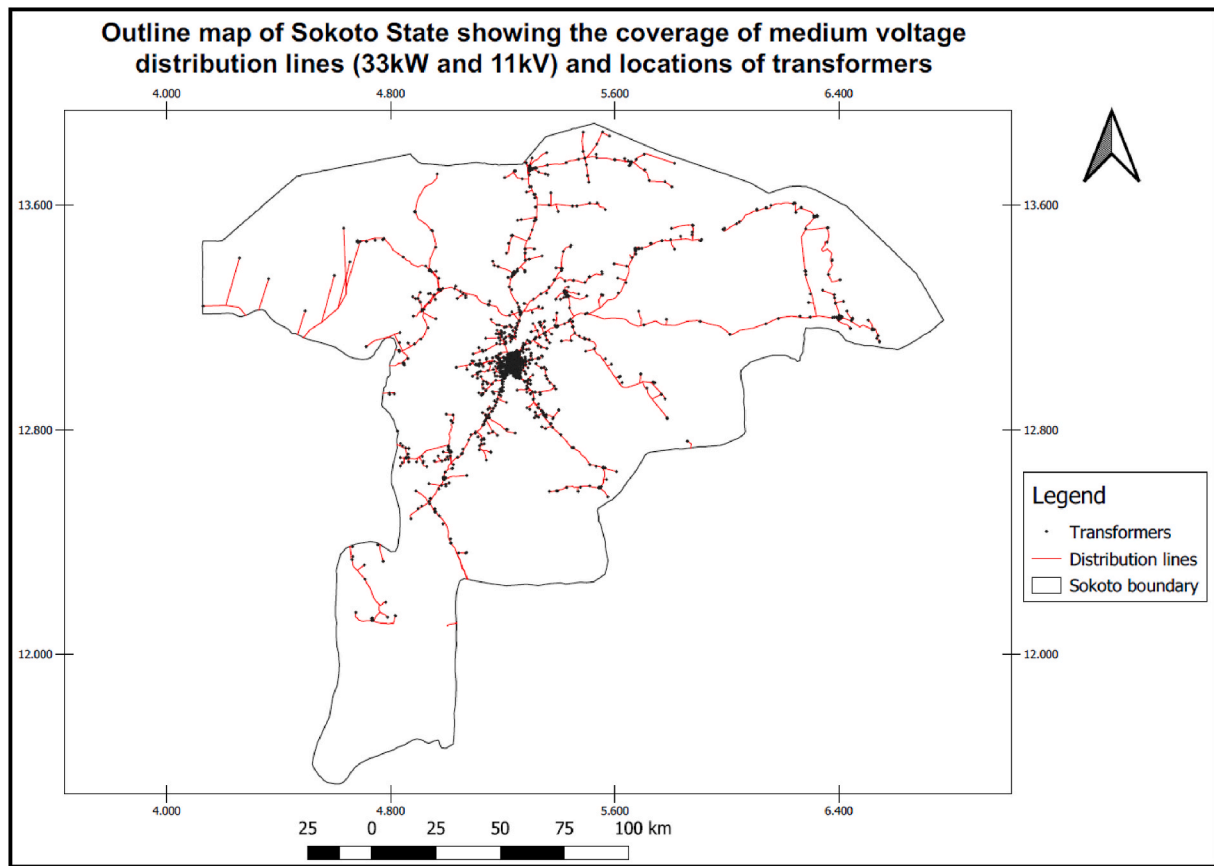


Fig. 3. Outline map of Sokoto State showing the coverage of the distribution lines (11 kV and 33 kV) and the location of the distribution transformers.

attributed to a community where electrification would provide access to electricity for 1,044 people, as well as 3 schools, 3 health facilities, and 2 markets. The cost of electrifying the community is US\$0.193 million. The expected socio-economic benefits for other unelectrified settlements ranked first to fifth in the MAUT analysis is presented in Table 10.

Furthermore, we present the map showing the top twenty communities that should be considered for electrification in the base-case scenario in Fig. 6 based on the result from MAUT.

In Table 11, we present the attributes of the unelectrified settlements that ranks first in each scenario. From Table 11, we observe that two unelectrified settlements rank top in all the scenarios. This implies that these communities should first be considered for electrification using the appropriate least-cost technology before others.

3.2.1. Further discussions

Given the significance of electricity access to socio-economic development in rural areas, electrifying the communities that have higher utility scores will yield higher net socio-economic benefits. The MAUT may also be used to provide answers to other questions. For example, given a specific budget, say US\$5 million, which communities should be electrified to ensure that the highest number of health facilities, educational facilities, number of persons, etc. have access to electricity? In the MAUT analysis in this study, we assume that the marginal utility functions (MUFs) of each criterion is linear. It is possible that these MUFs are not linear. For example, for a criterion like education, it seems realistic to assume that the MUF has a positive slope whose gradient decreases as the normalized performance score increases. That is, the utility when the number of schools that has electricity increases from 0 to 1 is expected to be higher than the utility when the electrified schools increase from 4 to 5. The assumption of linearity in the marginal utility function may affect the outcome of this study.

It is important to mention that beyond the criteria included in this

analysis, decision-makers have several other considerations that may be difficult to model in an analysis like this. For example, we observe from Fig. 6 that the top twenty unelectrified settlements that should be prioritized for electrification based on utility scores in the business-as-usual scenario are concentrated in certain areas, while other areas have no settlements in the top 20. However, the requirement of expending public funds in Nigeria is such that it allows for a fair geographic spread across the local government areas in a state. It is possible that the settlements with the top 20 utility scores in the other scenarios are also unevenly spread across the state. This is one of the challenges of implementing such a ranking system. This could raise issues around equity and fairness, as well as political tensions in the process of providing electricity access. One way to overcome this challenge is to filter the settlements by local government areas (LGAs⁴) and carry out the analysis at the LGA level. Another point to note is that the utility scores of the settlements are influenced by the weights assigned to the criteria in the different scenarios. If the weights are changed, the utility scores may change considerably. The weights used in the MAUT are usually subject to the preferences and considerations of the decision-maker.

There are financial, technical, governance and institutional barriers that need to be considered in the promotion and utilization of renewable energy technologies to achieve SDG7 [46]; [47,48]. Technical and institutional barriers are being addressed systematically by the Rural Electrification Agency in partnership with the regulator (i.e. the Nigeria Electricity Regulatory Commission). For example, technical codes for mini-grid projects have been developed alongside guidelines for

⁴ Local government areas are the 3rd administrative levels in Nigeria. Each state is divided into several local government areas and there are a total of 774 LGAs in Nigeria.

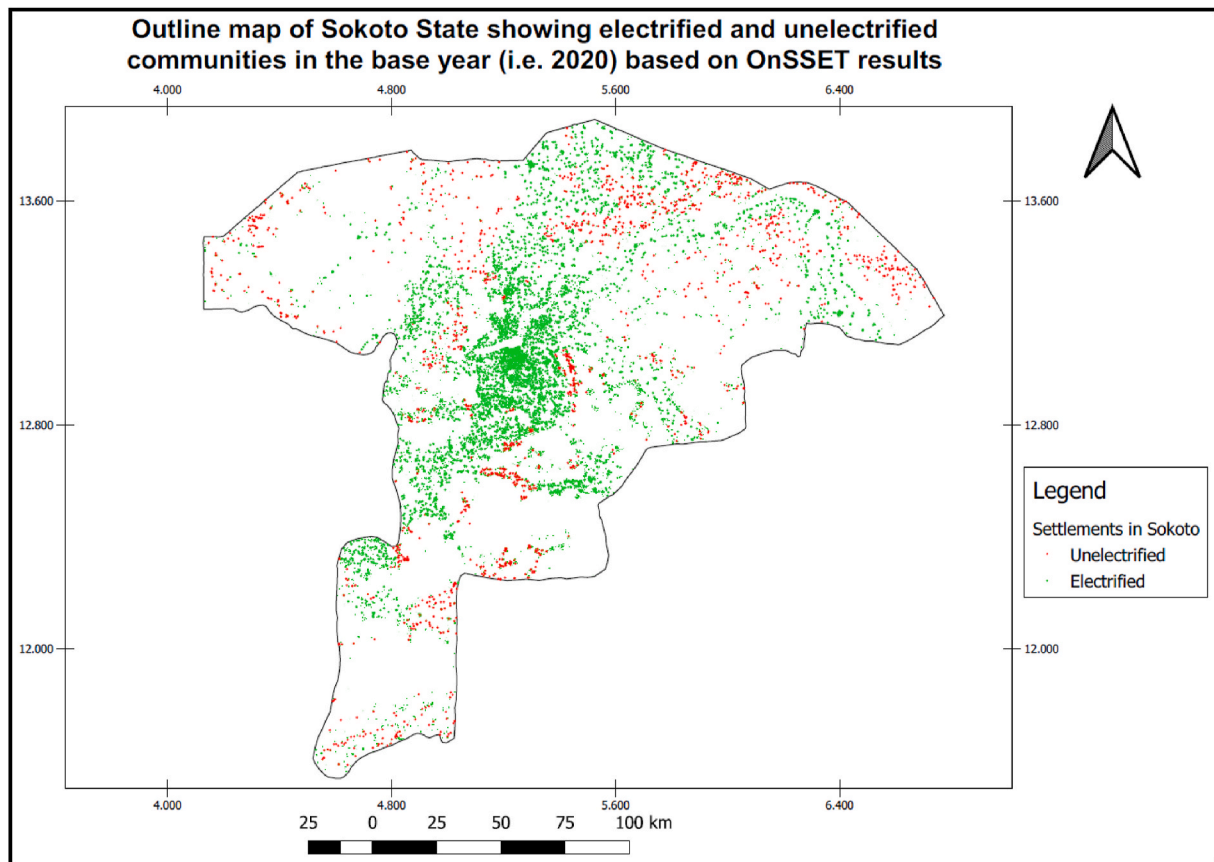


Fig. 4. Outline map of Sokoto State showing electrified and unelectrified settlements.

Table 4

Summary of information on population and electrification status of settlements for the base year 2020.

Item Description	Value
Total number of settlements	4,831
Minimum population	1
25th percentile	132
Median population	348
75th percentile	1,124
Max population	801,464
Average population	1,265.60
Total population	6,114,137
Number of unelectrified settlements	1,453
Number of electrified settlements	3,378
Population of unelectrified settlements	1,188,973
Population of electrified settlements	4,925,164

environment and social impact assessments. Furthermore, the exclusivity given to DisCos to distribute electricity in their respective coverage areas has been reconsidered to give room for private sector involvement in rural electrification.

However, the financial barriers, such as high investment cost per capita, lack of access to finance, and high investment risks [46] persist. The electricity sector in Nigeria is unbundled and the distribution segment is managed by the private sector which hitherto had exclusive right to distribute electricity in their respective geographic areas. Consequently, utility companies usually focus only on areas where cost recovery and profit are possible which often do not include these dispersed unelectrified settlements. In recent times, the World Bank and the African Development Bank have provided funding in form of grants to private sector entities to support investment in electricity access projects in the form of pro-poor public-private partnerships. These

experimental projects are still ongoing and the actual impact, challenges, and lessons will be known in due course. There are also some social enterprises that provide loans to support private sector investments in electricity access projects.⁵ The terms of such loans are usually better than commercial loans,⁶ however, the terms of the loans are still substantial and thus affect the viability of renewable energy projects.

4. Conclusion and policy recommendations

Off-grid technology options will play an important role to help the Government of Nigeria achieve its target of universal access to electricity by 2030. In this study, we determined the least-cost technology options, investment requirement, and capacity needed for electrifying all unelectrified communities in Sokoto State, using OnSSET. Our results show that mini-grid solar PV will be the least-cost technology option for electrifying most unelectrified settlements. By identifying the least-cost technology options for all the unelectrified communities, we provide a reasonable indicator of potential projects that could be technoeconomically viable. Given that large number of unelectrified settlements in the study area, we highlight the need to prioritize the settlements for electrification so as to have the highest net socio-economic benefits, and to allow for long-term multi-year electrification planning. We used the multi-attribute utility theory to prioritize settlements for electrification projects using several criteria and identified the settlements that will yield the maximum socio-economic benefits under different scenarios.

⁵ An example of social enterprise is All-on (<https://www.all-on.com/>).

⁶ The loans given by social enterprises is at an interest rate of 9 % while rates of loans given by commercial banks is often between 20 and 25%.

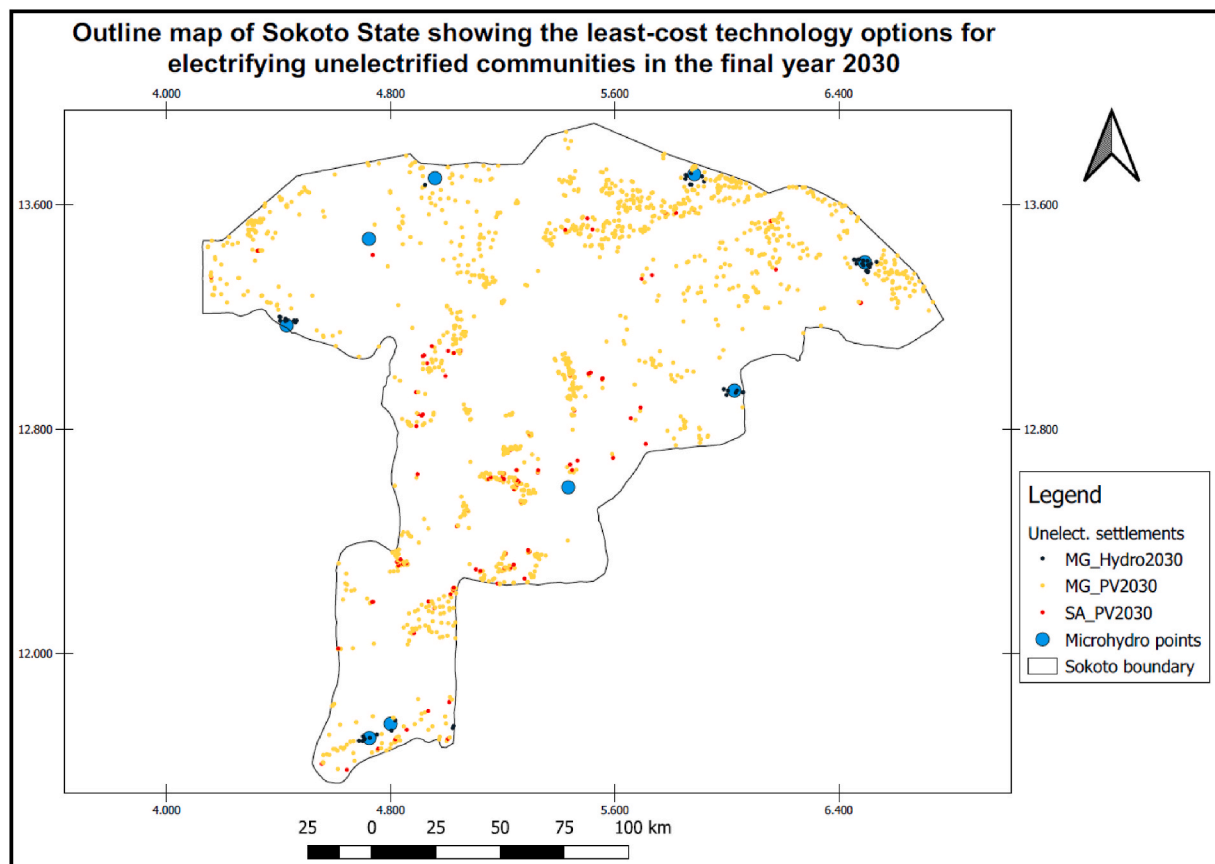


Fig. 5. Least-cost technology options for electrifying unelectrified settlements in the study area by 2030.

Table 5

Number of settlements where each technology is the least-cost option (disaggregated by population of settlements).

Settlements with:	Least-cost technology options/number of settlements			Total
	SA_PV2030	MG_PV2030	MG_Hydro2030	
Population ≤ 100	98	204	8	310
Population > 100 and ≤ 500	1	562	23	586
Population > 500 and ≤ 1000	0	197	9	206
Population > 1000	0	341	10	351
Total	99	1,304	50	1,453

Table 6

Projected 2030 population of settlements where each technology is the least-cost option (disaggregated by population of settlements).

Settlements with:	Least-cost technology option/population			Total
	SA_PV2030	MG_PV2030	MG_Hydro2030	
Population ≤ 100	2,207	14,503	468	17,179
Population > 100 and ≤ 500	172	148,398	6,823	155,394
Population > 500 and ≤ 1000	0	149,245	6,545	155,789
Population > 1000	0	930,669	19,783	950,452
Total	2,379	1,242,815	33,620	1,278,814

We have highlighted the barriers to the achievement of SDG7 and have noted that technical, institutional, governance barriers are being addressed. However, financial barriers still persist. The rural

Table 7

Additional capacity (kW) required to electrify settlements where each technology is the least-cost option (disaggregated by the population of settlements).

	SA_PV2030	MG_PV2030	MG_Hydro2030	Total
Population ≤ 100	198	1,444	23	1,665
Population > 100 and ≤ 500	15	14,671	333	15,020
Population > 500 and ≤ 1000	0	14,739	320	15,059
Population > 1000	0	92,154	966	93,120
Total	213	123,009	1,641	124,864

Table 8

Investment requirements (in million US\$) to achieve universal electricity access in the study area by 2030 (disaggregated by the population of settlements).

	SA_PV2030	MG_PV2030	MG_Hydro2030	Total
Population ≤ 100	0.529	3.284	0.066	3.879
Population > 100 and ≤ 500	0.041	28.353	0.645	29.040
Population > 500 and ≤ 1000	–	27.239	0.574	27.813
Population > 1000	–	168.971	1.711	170.683
Total	0.570	227.848	2.997	231.414

electrification sub-sector in Nigeria is now open to the private sector or social enterprises especially under the Nigeria Electrification Project which is funded by the African Development Bank and the World Bank. These multilateral development banks provide incentives to the private sector in the form of grants to improve the financial viability of rural electrification projects. However, given the large market for renewable energy equipment which can be used for mini-grid electrification to

Table 9

Investment per capita (US\$/person) required to achieve universal electricity access in the study area by 2030 (disaggregated by the population of settlements).

	SA_PV2030	MG_PV2030	MG_Hydro2030
Population ≤100	239.40	229.74	147.92
Population >100 and ≤500	236.87	193.96	97.25
Population >500 and ≤1000	–	182.55	87.87
Population >1000	–	181.33	86.34

achieve universal electrification, it is important that incentives be given to the private sector to manufacture and/or assemble renewable energy equipment in Nigeria. This will substantially reduce the cost of mini-grid projects as well as the associated risks of such projects in terms of cost recovery due to exchange rate fluctuations.

This research is highly replicable and can be expanded to cover the whole of Nigeria or any other country, subject to availability of data on coverage of the existing MV or LV distribution lines, locations of transformers, or other critical datasets required by OnSSET. Analysing other states can provide additional insight on how different local-level

socio-economic variables can affect decision making on electrification projects. We recommend that additional investment be made, not just on the collection of these datasets, but also on making these datasets readily available to researchers and decision makers.

CRediT authorship contribution statement

Salisu Isihak: Conceptualization, Methodology, Investigation, Validation, Formal analysis, Writing – original draft. **Uduak Akpan:** Writing – review & editing. **Subhes Bhattacharyya:** Writing – review & editing.

U4RIA compliance

The use of OnSSET, which is an open-source energy planning tool, as well as the spatial data used in the MAUT align with the recent trend that advocates the use of open-source tools to carry out energy planning and energy system modelling within the U4RIA framework principles for energy modelling for policy support (EMoPS) [49–51]. The U4RIA requires that data used use for energy planning and energy modelling be

Table 10

Attributes of the top five unelectrified settlements that should be considered for electricity access projects based on the base-case scenario of the MAUT.

Rank	1D	Population of settlement	Least cost technology option	Investment Cost (million US\$)	No. of polling units	No. of schools	No. of markets	No. of health facilities
1	572	1,044	Mini-grid PV	0.193	4	3	1	3
2	580	1,905	Mini-grid PV	0.344	4	4	0	3
3	1,048	1,250	Mini-grid PV	0.225	5	3	0	3
4	570	4,015	Mini-grid PV	0.737	3	4	1	2
5	1,046	629	Mini-grid PV	0.117	4	2	0	3

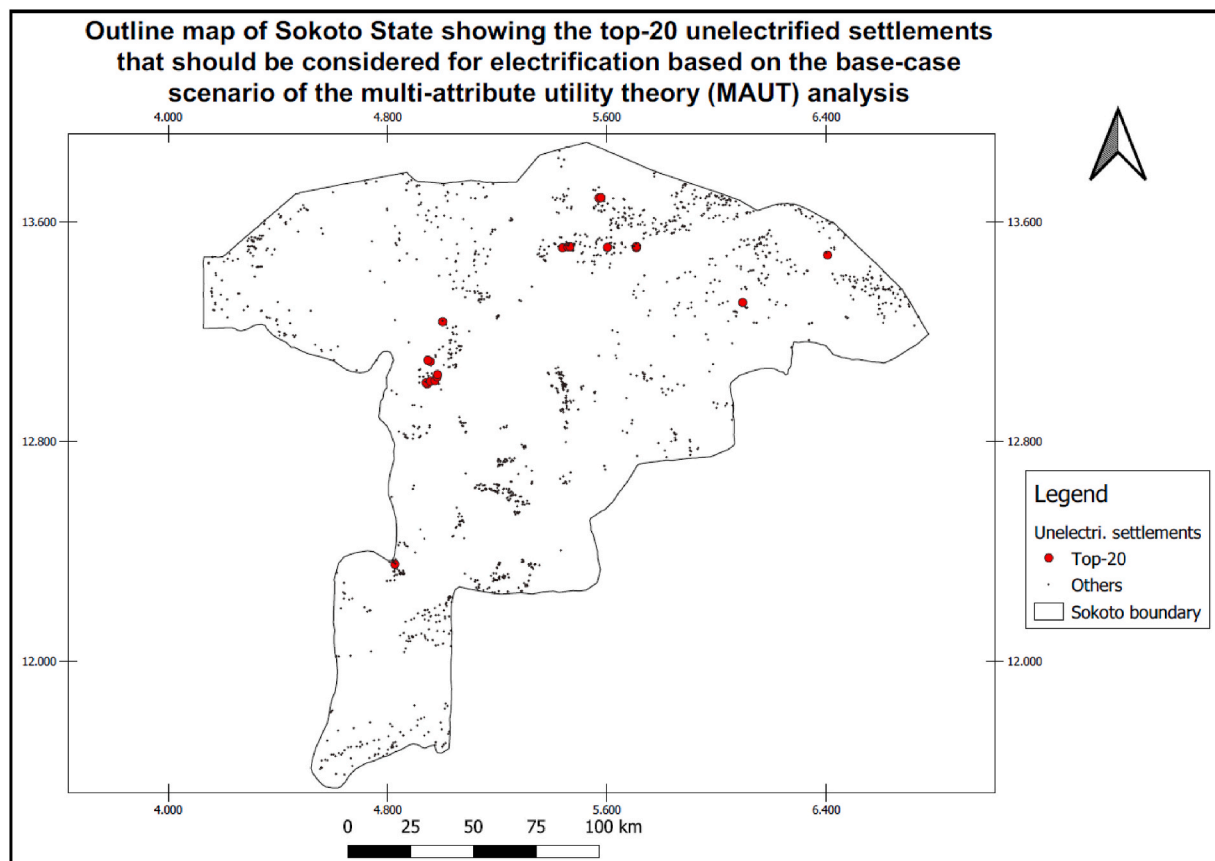


Fig. 6. Outline map of Sokoto State showing the top-twenty unelectrified settlements that should be considered for electricity access projects based on the base-case scenario of the multi-attribute utility theory (MAUT).

Table 11
Attributes of the unelectrified settlement that ranks first in each scenario in the MAUT.

Scenarios	ID	Least Cost tech.	2030Pop est.	Investment Cost (2030) (million US\$)	No. of polling units	No. of schools	No. of markets	No. of health facilities	Global utility score
Base case	572	Mini-grid PV	1,044	0.193	4	3	1	3	0.766
Demographic	1,390	Mini-grid PV	6,589	1.179	8	4	0	2	0.500
Financial	572	Mini-grid PV	1,044	0.193	4	3	1	3	0.786
Social	572	Mini-grid PV	1,044	0.193	4	3	1	3	0.730
Economic	572	Mini-grid PV	1,044	0.193	4	3	1	3	0.810
Political	1,390	Mini-grid PV	6,589	1.179	8	4	0	2	0.748

“ubuntu” i.e. reflect the perspective of different stakeholders in the energy modelling community, retrievable, reusable, repeatable, reconstructable, interoperable, and auditable. The data for the OnSSET are already in the public domain on different websites. The datasets are provided as supplementary material. The dataset can be accessed, downloaded, viewed (retrievability), reused (reusability), and the steps taken in the modelling process can be repeated (repeatability) or reconstructed (reconstructability). The GIS shapefiles and raster data are in formats that are interoperable. Hence, the entire modelling process is auditable.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.esr.2025.101807>.

References

- [1] IEA, SDG7: data and projections, Access to electricity (2024). <https://www.iea.org/reports/sdg7-data-and-projections/access-to-electricity>. (Accessed 10 January 2025).
- [2] IEA, Tracking SDG7: the energy progress report 2023 [Online] Available at: <https://iea.blob.core.windows.net/assets/9b89065a-cb4-404c-a53e-084982768baf/SDG7-Report2023-FullReport.pdf>, 2023. (Accessed 10 January 2025).
- [3] O.D. Odou, R. Bhandari, R. Adamou, Hybrid off-grid renewable power system for sustainable rural electrification in Benin, *Renew. Energy* 145 (2020) 1266–1279, <https://doi.org/10.1016/j.renene.2019.06.032>.
- [4] F. Yang, M. Yang, Rural electrification in sub-Saharan Africa with innovative energy policy and new financing models, *Mitig. Strategies Glob. Change* 23 (2018) 933–952, <https://doi.org/10.1007/s11027-017-9766-8>.
- [5] U. Akpan, Technology options for increasing electricity access in areas with low electricity access rate in Nigeria, *Soc. Econ. Plann. Sci.* 51 (2015) 1–12.
- [6] S. Isihak, U. Akpan, S. Ohiare, The evolution of rural household electricity demand in grid-connected communities in developing countries: result of a survey, *Future Cities and Environment* 10 (6) (2020) 1–11.
- [7] Rural Electrification Agency, Rural electrification strategy and plan (RESIP), in: Federal Ministry of Power, Works and Housing, 2016. Retrieved July 20, 2022.
- [8] FGN, National Renewable Energy and Energy Efficiency Policy, Federal Ministry of Power, 2015.
- [9] O.O. Ajayi, The potential for wind energy in Nigeria, *Wind Eng.* 34 (3) (2010) 303–311, <https://doi.org/10.1260/0309-524X.34.3.303>.
- [10] A.G. Dagnachew, P.L. Lucas, A.F. Hof, D. Gernaat, H.-S. de Boer, D.P. van Vuuren, The role of decentralized systems in providing universal electricity access in Sub-Saharan Africa – a model-based approach, *Energy* 139 (2017) 184–195.
- [11] S.O. Oyedepo, On energy for sustainable development in Nigeria, *Renew. Sustain. Energy Rev.* 16 (5) (2012) 2583–2598.
- [12] F. Kemausuor, E. Adkins, I. Adu-Poku, A. Brew-Hammond, V. Modi, Electrification planning using network planner tool: the case of Ghana, *Energy Sustain. Dev.* 19 (2014) 92–101.
- [13] A. Korkovelos, B. Khavari, A. Sahlberg, M. Howells, C. Arderne, The role of open access data in geospatial electrification planning and the achievement of SDG7. An OnSSET-Based case study for Malawi, *Energies* 12 (2019), <https://doi.org/10.3390/en12071395>.
- [14] C. Berndtsson, Open Geospatial Data for Energy Planning, KTH School of Industrial Engineering and Management, Stockholm, 2016. Retrieved from, <https://www.diva-portal.org/smash/get/diva2:927179/FULLTEXT02.pdf>.
- [15] B. Khavari, A. Sahlberg, Geo-Spatial Electricity Demand Assessment & Hybrid off-grid Solutions to Support Electrification Efforts Using Onset: the Case Study of Tanzania, KTH School of Industrial Engineering and Management, 2017. Retrieved from, <http://www.diva-portal.org/smash/get/diva2:1197546/FULLTEXT01.pdf>.
- [16] A.L. Temesgen, Y.T. Wassie, E.O. Ahlgren, Analyzing grid extension suitability: a case study of Ethiopia using OnSSET, *Energy Strategy Rev.* 52 (2024) 101292.
- [17] S. Ohiare, Expanding electricity access to all in Nigeria: a spatial planning and cost analysis, *Energy Sustain. Soc.* (2015) 5–8, <https://doi.org/10.1186/s13705-015-0037-9>.
- [18] D. Mentis, M. Welsch, F.F. Nerini, O. Broad, M. Howells, M. Bazilian, H. Rogner, A GIS-based approach for electrification planning—A case study on Nigeria, *Energy Sustain. Dev.* (2015) 142–150, <https://doi.org/10.1016/j.esd.2015.09.007>.
- [19] World Bank, Achieving Universal Access in the Kaduna Electricity Service Area, World Bank, 2016. Retrieved from, <http://documents.worldbank.org/curated/en/782491487138851242/pdf/112802-Report-Kaduna-Electric.pdf>.
- [20] S. Isihak, U. Akpan, S. Bhattacharyya, Evolution of GIS-based rural electrification planning models and an application of OnSSET in Nigeria, *Renew. Sustain. Energy Transit.* 2 (2022) 100019.
- [21] S.R. Isihak, Achieving universal electricity access in line with SDG7 using GIS-based model: an application of OnSSET for rural electrification planning in Nigeria, *Energy Strategy Rev.* 45 (2023) 101021.
- [22] A. Kumar, B. Sah, A.R. Singh, Y. Deng, X. He, P. Kumar, R. Bansal, A review of multi criteria decision making (MCDM) towards sustainable renewable energy development, *Renew. Sustain. Energy Rev.* 69 (2017) 596–609.
- [23] J.A. Cherni, I. Dynner, F. Henao, P. Jaramillo, R. Smith, R.I. Font, Energy supply for sustainable rural livelihoods: a multi-criteria decision-support system, *Energy Policy* 35 (2007) 1493–1504.
- [24] T. Buchholz, E. Rametsteiner, T.A. Volk, V.A. Luzadis, Multi-criteria analysis for bioenergy systems assessments, *Energy Policy* 37 (2009) 484–495.
- [25] A. Kumar, Y. Deng, X. He, P. Kumar, A multi criteria decision based rural electrification system, in: IECON 2016 - 42nd Annual Conference of the IEEE Industrial Electronics Society, Florence, Italy, 2016.
- [26] J.C. Rojas-Zerpa, J.M. Yusta, Application of multicriteria decision methods for electric supply planning in rural and remote areas, *Renew. Sustain. Energy Rev.* 32 (2015) 557–571.
- [27] M. Amer, T.U. Daim, Selection of renewable energy technologies for a developing county: a case of Pakistan, *Energy Sustain. Dev.* 15 (4) (2011) 420–435.

- [28] S. Ahmad, R.M. Tahar, Selection of renewable energy sources for sustainable development of electricity generation system using analytic hierarchy process: a case of Malaysia, *Renew. Energy* 63 (2014) 458–466.
- [29] M. Uyan, GIS-based solar farms site selection using analytic hierarchy process (AHP) in Karapinar region, Konya/Turkey, *Renew. Sustain. Energy Rev.* w8 (2013) 11–17.
- [30] M. Tahri, M. Hakdaoui, M. Maanan, The evaluation of solar farm locations applying geographic information system and multi-criteria decision-making methods: case study in southern Morocco, *Renew. Sustain. Energy Rev.* 15 (2015) 1354–1362.
- [31] S. Pohekar, M. Ramachandran, Application of multi-criteria decision making to sustainable energy planning—A review, *Renew. Sustain. Energy Rev.* 8 (2004) 365–381.
- [32] J.-J. Wang, Y.-Y. Jing, C.-F. Zhang, J.-H. Zhao, Review on multi-criteria decision analysis aid in sustainable energy decision-making, *Renew. Sustain. Energy Rev.* 13 (2009) 2263–2278.
- [33] I. Siksnelyte, E.K. Zavadskas, D. Streimikiene, D. Sharma, An overview of multi-criteria decision-making methods in dealing with sustainable energy development issues, *Energies* 11 (2018) 2754, <https://doi.org/10.3390/en1102754>.
- [34] E. Ilbahar, S. Cebi, C. Kahraman, A state-of-the-art review on multi-attribute renewable energy decision making, *Energy Strategy Rev.* 25 (2019) 18–33.
- [35] C. Paredes-Budon, L. Reyes-Chamorro, Prioritization of isolated communities for the development of renewable-energy projects in wide geographical areas, in: *IEEE PES Transmission & Distribution Conference and Exhibition - Latin America (T&D LA)*, IEEE, Montevideo, Uruguay, 2020, <https://doi.org/10.1109/TDLA47668.2020.9326222>.
- [36] D.F. Gómez-Hernández, B. Domenech, M. Juanpera, L. Ferrer-Martí, Ranking projects in regional electrification plans considering technical and social criteria. Case study in Mexico, *Energy Sustain. Dev.* 77 (2023) 101336.
- [37] M. Juanpera, P. Blechinger, L. Ferrer-Martí, M. Hoffmann, R. Pastor, Multicriteria-based methodology for the design of rural electrification systems. A case study in Nigeria, *Renew. Sustain. Energy Rev.* 133 (2020) 110243, <https://doi.org/10.1016/j.rser.2020.110243>.
- [38] M. Juanpera, L. Ferrer-Martí, R. Pastor, Multi-stage optimization of rural electrification planning at regional level considering multiple criteria. Case study in Nigeria, *Appl. Energy* 314 (2022) 118926, <https://doi.org/10.1016/j.apenergy.2022.118926>.
- [39] NPC, National Population Commission, Nigeria: Demographic and Health Survey 2013, National Population Commission, Nigeria, 2013, 2014, June. Retrieved August 12, 2022, from, <https://ngfrepository.org.ng:8443/bitstream/123456789/3139/1/NDHS%202013.pdf>.
- [40] D. Mentis, M. Howells, H. Rogner, A. Korkovelos, C. Arderne, E. Zepede, E. Scholtz, Lighting the world: the first application of an open source spatial electrification tool (OnSSET) on Sub-Saharan Africa, *Environmental Research Letters* (2017), <https://doi.org/10.1088/1748-9326/aa7b29>.
- [41] U. Akpan, R. Morimoto, An application of multi-attribute utility theory (MAUT) to the prioritization of rural roads to improve rural accessibility in Nigeria, *Soc. Econ. Plann. Sci.* (2022) 101256, <https://doi.org/10.1016/j.seps.2022.101256>.
- [42] T. Saaty, *The Analytic Hierarchy Process*, McGraw-Hill, New York, 1980.
- [43] R.W. Saaty, The analytic hierarchy process-what it is and how it is used, *Math. Model.* 9 (3–5) (1987) 161–176.
- [44] A. Ishizaka, P. Nemery, *Multicriteria Decision Analysis: Methods and Software*, first ed., John Wiley & Sons, Ltd, West Sussex, 2013.
- [45] ESMAP, Beyond Connections: Energy Access Redefined, S.L, World Bank, 2015 [Online] Available at: <https://openknowledge.worldbank.org/server/api/core/bitstreams/248a7205-e926-5946-9025-605b8035ad95/content>. (Accessed 15 January 2025).
- [46] A.G. Dagnachew, A.F. Hof, M.R. Roelfsema, D.P. Vuuren, Actors and governance in the transition toward universal electricity access in Sub-Saharan Africa, *Energy Policy* 143 (2020) 111572, <https://doi.org/10.1016/j.enpol.2020.111572>.
- [47] M. Yaqoot, P. Diwan, T.C. Kandpal, Review of barriers to the dissemination of decentralized renewable energy systems, *Renew. Sustain. Energy Rev.* 58 (2016) 477–490, <https://doi.org/10.1016/j.rser.2015.12.224>.
- [48] G. Falchetta, A.G. Dagnachew, A.F. Hof, D.J. Milne, The role of regulatory, market and governance risk for electricity access investment in sub-Saharan Africa, *Energy Sustain. Dev.* 62 (2021) 136–150, <https://doi.org/10.1016/j.esd.2021.04.002>.
- [49] M. Howells, J. Quiros-Tortos, R. Morrison, H. Rogner, W. Blyth, G. Godínez, et al., *Energy system analytics and good governance-U4RIA goals of Energy Modelling for Policy Support*, 2021.
- [50] L. Dixon, R. Yeganyan, C. Cannone, M. Howells, V. Foster, F. Plazas-Niño, Techno-economic data and assumptions for open energy modelling of decarbonisation pathways in the Philippines, *Data in Brief* 54 (2024) 110459.
- [51] F. Plazas-Niño, N. Tan, M. Howells, V. Foster, J. Quiros-Tortós, Uncovering the applications, developments, and future research directions of the open-source energy modelling system (OSeMOSYS): a systematic literature review, *Energy Sustain. Dev.* 85 (2025) 101629.